SIMATS ENGINEERING

ASSESSMENT TOOL 2

SCENARIO-BASED QUESTIONS

Course Name	COMPUTER VISION
Course Code	ITA05
Course Outcome	CO2 – Apply image enhancement and segmentation techniques for extracting meaningful regions from images (BTL 3)
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Programme	B.Tech (Information Technology)
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Assessment	Assessment Tool 2 – Scenario-Based Questions
Maximum Marks	50

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CODE OF CONDUCT

I, G. Harendra (Register Number: 192521215), certify that this submission is my original work and that I have adhered to the academic guidelines specified for this assessment. I understand that any violation of academic integrity rules may result in disciplinary action.

Student Signature: _____

Date: _____

1. Dental X-Ray Enhancement and Cavity Region Extraction

Scenario and Objective

A dental clinic uses a digital radiography system to help dentists locate early-stage cavities (caries) in teeth. The X-ray images are dark and low in contrast, with quantum noise and overlapping tooth structures, so small cavity regions blend into the surrounding enamel and dentine.

Processing objective: Improve the visibility of subtle cavity-like regions, suppress noise without destroying fine boundaries, and outline candidate regions for dentist review.

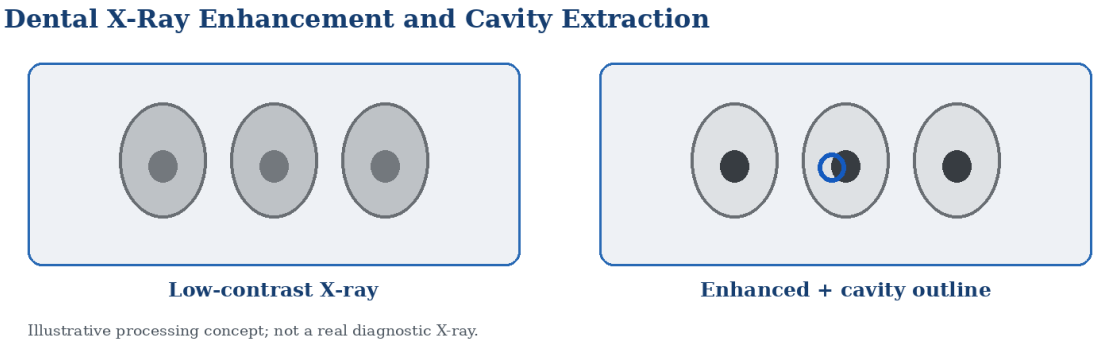


Figure: Illustrative computer-vision concept for the given scenario.

Proposed Processing Sequence

Step 1 – Input and inspection: Load the grayscale dental X-ray and inspect its intensity range, contrast, noise level and possible saturation. Preserve the original image so the enhanced result can be compared with it.

Step 2 – Grey-level transformation: Apply a logarithmic transformation when darker intensity details need expansion, or a power-law (gamma) transformation with an appropriate gamma value to brighten a dark image. A negative transformation can be used selectively when reversing intensity makes a feature easier to inspect; it is not the primary choice for every X-ray.

Step 3 – Histogram enhancement: Use histogram equalization to redistribute intensity values and improve global contrast. If global equalization over-amplifies noise or produces uneven contrast, use a local contrast method such as CLAHE with a conservative clip limit.

Step 4 – Noise reduction: Apply a small-window median filter to suppress impulse-like noise while preserving edges better than a broad averaging filter. If the noise is approximately Gaussian, a mild Gaussian filter may be considered. Keep the window small so thin cavity boundaries are not blurred.

Step 5 – Region extraction: Use adaptive thresholding when local brightness varies, or seed-based region growing when a reliable seed point and similarity criterion are available. Exclude tiny isolated components and inspect candidate regions in the context of the surrounding tooth anatomy.

Step 6 – Edge and boundary detection: Apply a Canny edge detector to identify candidate boundaries, then use morphological closing or edge linking to bridge short gaps. Overlay the outline on the enhanced X-ray for visual verification.

Step 7 – Validation: Compare the original, enhanced, segmented and outlined images. The result is a decision-support aid only: a qualified dentist must confirm whether a candidate region represents caries.



Figure: Processing pipeline designed for the scenario.

Summary

Log/gamma correction improves dark-detail visibility; histogram equalization improves contrast; a small median filter reduces noise; adaptive

thresholding or region growing extracts candidates; and Canny plus edge linking outlines them.

Expected Output

An enhanced dental X-ray with suspected cavity candidates outlined for expert review.

2. Night-Time Vehicle Number Plate Region Extraction

Scenario and Objective

A traffic police department installs CCTV cameras at a toll gate to capture vehicles at night. The images suffer from very low brightness, headlight glare, sensor noise and motion blur. The system must locate and extract the number-plate region so the characters can be read in a later stage.

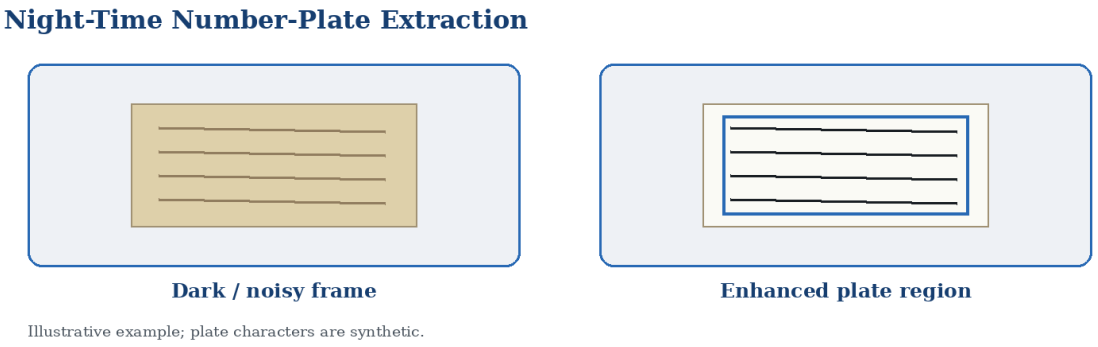


Figure: Illustrative computer-vision concept for the given scenario.

Proposed Processing Sequence

Step 1 – Input and brightness analysis: Read the night-time frame and identify underexposed areas, glare, motion blur and sensor noise. Avoid treating clipped headlight regions as recoverable detail because saturated pixels may have lost information.

Step 2 – Brightness and histogram correction: Apply a power-law transformation to lift dark regions, followed by histogram equalization or CLAHE to improve local contrast. Use conservative settings and, where possible, a glare mask so bright headlights are not amplified further.

Step 3 – Noise removal and sharpening: Use a median filter to suppress isolated sensor noise. Apply mild unsharp masking or Laplacian-based sharpening to strengthen plate-character edges. Sharpening cannot fully reverse severe motion blur, so avoid excessive settings that create halos.

Step 4 – Edge detection: Use Canny edge detection to locate strong intensity transitions around the plate and characters. Apply morphological closing to connect short edge gaps caused by noise or weak illumination.

Step 5 – Line and rectangle detection: Use a Hough line transform or contour analysis to find approximately parallel horizontal and vertical boundaries. Rank rectangular candidates using aspect ratio, size, edge density and position within the vehicle region.

Step 6 – Plate extraction and character separation: Crop the most plausible plate rectangle, correct its perspective if tilted, and convert it to grayscale. Apply adaptive thresholding to separate dark characters from a lighter plate background, or invert the binary result if required by the plate polarity.

Step 7 – Output and validation: Return the cropped plate region and a binary character image for later optical character recognition. Check that the complete plate is included and that glare, blur or thresholding has not erased character strokes.

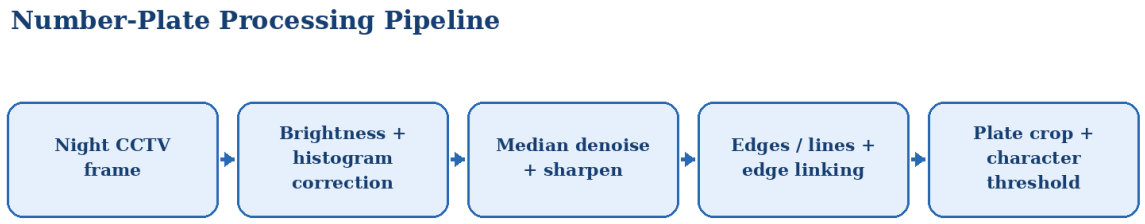


Figure: Processing pipeline designed for the scenario.

Summary

Brightness correction reveals the plate; median filtering reduces noise; sharpening strengthens strokes; Canny and line detection locate the rectangle; and adaptive thresholding separates characters.

Expected Output

A cropped number-plate region and a thresholded character image ready for a later OCR stage.

3. Textile Fabric Defect Detection Using Frequency-Domain Processing

Scenario and Objective

A textile mill inspects woven fabric rolls using a line-scan camera. The fabric has a regular, repeating weave pattern, and defects such as holes, broken threads, oil stains and knots disturb this pattern. The captured images also contain periodic noise from machine lighting and uneven illumination across the roll.

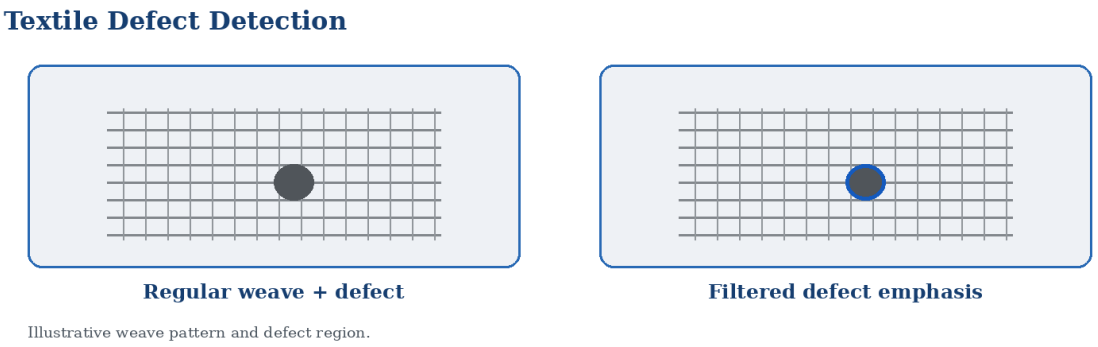


Figure: Illustrative computer-vision concept for the given scenario.

Proposed Processing Sequence

Step 1 – Image preparation: Convert the fabric image to grayscale if colour is not needed for the defect class. Correct uneven illumination using background estimation or local histogram processing so slow brightness changes do not dominate defect detection.

Step 2 – Transform to the frequency domain: Compute the two-dimensional Fast Fourier Transform (FFT) and display the log magnitude spectrum. Repeating weave texture and periodic lighting noise may appear as structured peaks or bands away from the spectrum centre.

Step 3 – Select a frequency filter: Use notch-reject filters to suppress identifiable periodic-noise peaks. A band-reject filter is appropriate when unwanted energy occupies a known frequency band. A low-pass filter can suppress high-frequency noise but may blur small defects; a high-pass filter can highlight abrupt defects but also amplify noise.

Step 4 – Inverse transform: Apply the selected filter mask to the Fourier spectrum and perform the inverse FFT to reconstruct the filtered image. Inspect the result for ringing, loss of thin threads or accidental removal of defect information.

Step 5 – Contrast and segmentation: Correct residual illumination using histogram equalization or CLAHE. Apply a suitable threshold to create a defect candidate mask, choosing adaptive thresholding if the fabric brightness varies spatially.

Step 6 – Boundary extraction: Use connected-component analysis and boundary detection to outline candidate holes, stains, knots or broken-thread regions. Remove components that are too small or inconsistent with expected defect shapes.

Step 7 – Why frequency-domain processing?: Periodic patterns and lighting interference are often easier to isolate by their frequency peaks than by a local spatial filter. Spatial filters remain useful for denoising and cleanup, but a frequency-domain notch filter can target specific repeating interference while retaining other image content.

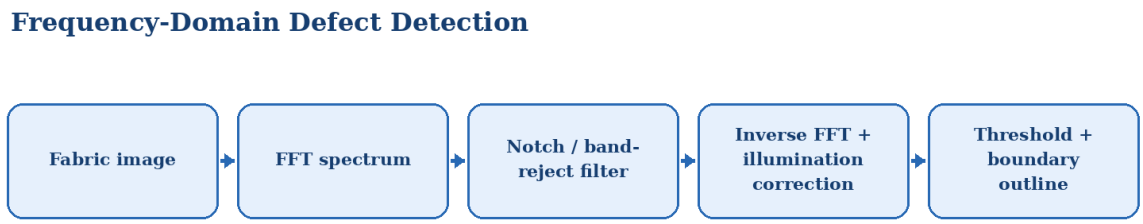


Figure: Processing pipeline designed for the scenario.

Summary

FFT exposes periodic weave and lighting components; notch/band-reject filtering targets those frequencies; inverse FFT reconstructs the image; then thresholding and boundary analysis isolate defects.

Expected Output

A filtered fabric image and a mask outlining candidate defects for inspection.

4. Blood Cell Segmentation in Microscopic Images

Scenario and Objective

A pathology laboratory uses an automated microscope to capture images of stained blood smears for counting red and white blood cells. The images show uneven staining, poor contrast between cells and background, touching or overlapping cells, and noise from dust particles on the slide.

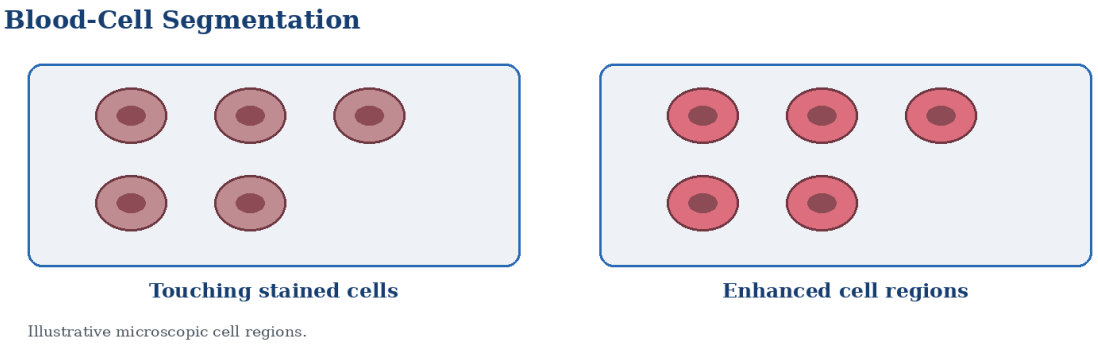


Figure: Illustrative computer-vision concept for the given scenario.

Proposed Processing Sequence

Step 1 – Input and illumination correction: Acquire a microscope image and inspect stain variation, background intensity, dust and cell overlap. Estimate and correct uneven background illumination before segmentation.

Step 2 – Contrast enhancement: Apply histogram equalization or CLAHE to increase the contrast between cells and the background. For colour images, use a suitable colour space and enhance the intensity channel rather than indiscriminately changing all colour channels.

Step 3 – Noise filtering: Use a small median filter to remove isolated particles while preserving cell boundaries. Choose filter size carefully because large filters can merge nearby cells or erase small cell details.

Step 4 – Thresholding: Apply global thresholding if cell and background intensities are consistently separated. Under uneven staining, use adaptive thresholding to calculate local thresholds and create an initial foreground mask.

Step 5 – Separate touching cells: Use region growing when seed points and intensity similarity are reliable. For touching, roughly round cells, a distance transform followed by watershed segmentation can help split connected components. Region splitting and merging is another option when regions can be divided by homogeneity criteria.

Step 6 – Edge and boundary detection: Apply Canny or a suitable gradient operator to obtain candidate cell edges. Use morphological operations and connected-component analysis to produce clean individual-cell outlines.

Step 7 – Quality checks: Overlay the boundaries on the original micrograph and inspect under-segmentation, over-segmentation and missed cells. Only after validation should the system calculate cell counts or measurements.

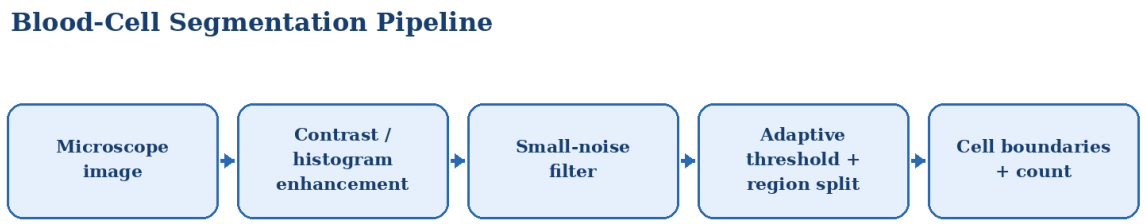


Figure: Processing pipeline designed for the scenario.

Summary

Contrast enhancement separates cell tones; adaptive thresholding creates a foreground mask; region-based splitting separates touching cells; and edge detection outlines each cell for counting.

Expected Output

A microscopic image with individual cell boundaries suitable for validated cell counting.

5. Text Extraction from Degraded Historical Documents

Scenario and Objective

A state archive is digitising old handwritten manuscripts. The scanned pages contain yellowed paper, ink bleed-through from the reverse side, stains, faded strokes and uneven lighting across the page. The archive requires clean extraction of text regions before handwriting recognition.

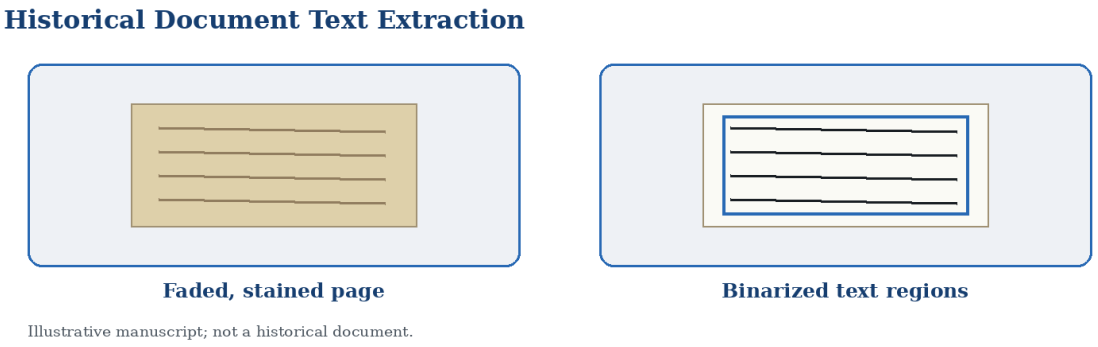


Figure: Illustrative computer-vision concept for the given scenario.

Proposed Processing Sequence

Step 1 – Input and grayscale conversion: Load the scanned page and preserve the colour original for archival use. Convert a working copy to grayscale or a suitable luminance channel, and estimate uneven background illumination.

Step 2 – Grey-level and histogram enhancement: Use a power-law transformation or contrast stretching to make faint ink more visible. Apply histogram equalization or CLAHE to improve local contrast between faded strokes and the paper, while monitoring whether stains and bleed-through become stronger.

Step 3 – Denoising and stroke enhancement: Use a mild median filter to remove small isolated specks. Apply conservative unsharp masking to strengthen pen strokes. Broad smoothing should be avoided because it may erase thin handwritten marks.

Step 4 – Global versus adaptive thresholding: Global thresholding, including Otsu's method, selects one threshold for the whole image and works best when foreground and background intensities are reasonably separable. Adaptive thresholding computes local thresholds and is more suitable here because page illumination, paper colour and ink density vary across the page.

Step 5 – Binary cleanup: Create a text mask using adaptive thresholding, then apply small morphological opening to remove isolated noise and closing to reconnect broken strokes. Choose structuring elements carefully so adjacent letters do not merge.

Step 6 – Text-line and word-region detection: Use connected components, horizontal projection profiles and boundary detection to estimate text lines and word regions. For connected handwriting, group nearby components using spacing, alignment and stroke continuity rather than relying only on individual character boxes.

Step 7 – Output and verification: Overlay text-region boxes on the scanned page and save the clean binary image for handwriting recognition. Compare with the original to ensure faded strokes are preserved and stains or bleed-through have not been mistaken for text.

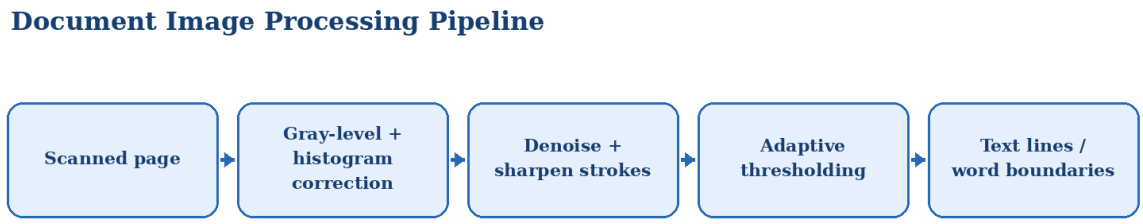


Figure: Processing pipeline designed for the scenario.

Summary

Adaptive thresholding is preferred because lighting and paper tone vary across the page. Local enhancement, careful denoising, stroke sharpening and boundary grouping prepare text regions for handwriting recognition.

Expected Output

A cleaned text-region mask and line/word-region boxes ready for handwriting-recognition processing.